

SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

MEDICAL ELECTRONICS
MODEL QUESTION PAPER – SET-1

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1	Define action potential	M 1.01	R
2	Mention cell repolarization	M1.01	R
3	List the types of blood cells	M2.01	R
4	Give the normal values of systolic and diastolic blood pressure	M2.02	R
5	List any two properties of Laser	M2.04	R
6	State the need of pacemaker	M3.01	R
7	State the use of respirators	M3.03	R
8	State the need of bio telemetry	M4.03	R
9	Define micro shock	M4.04	R

PART B

II. Answer any Eight questions from the following
(8 x 3= 24 Marks)

		Module Outcome	Cognitive level
1	Illustrate QRS wave form	M 1.02	U
2	Explain EMG	M1.04	U
3	Demonstrate the operation of surface electrode	M1-02	U
4	List the constituents and functions of WBC	M2.01	R
5	Describe pulse oxymeter	M2.03	U
6	Define fibrillation. How it could be treated.	M3.01	R

7	Classify the types of hemo dialysis	M3.02	U
8	Illustrate CT scanning	M4.01	U
9	Describe the working principle of Ultra sound scanning	M4.02	U
10	List the effects of electric current in human body	M4.04	R

PART C

Answer all questions from the following (6x 7 = 42 Marks)

		Module Outcome	Cognitive level
1	Give details on types of electrodes used in bio potential measurements	M1.02	U
OR			
2	Illustrate the block diagram of EEG machine	M1.03	U
3	Explain the indirect method of blood pressure measurement	M2.02	U
OR			
4	List the applications of Laser in medical field	M2.04	R
5	Classify the types of pace makers	M3.01	U
OR			
6	Illustrate Hemo dialysis	M3.02	U
7	Illustrate MRI scanning	M4.02	U
OR			
8	Elaborate Bio telemetry system with block diagram	M4.03	U
9	Explain Coulter counter method of blood cell counting	M2.01	U
OR			
10	Explain PCO2 analyzers	M2.03	U
11	Explain the classification of ventilators	M3.03	U
OR			
12	Illustrate DC defibrillators. Compare AC and DC defibrillators	M3.01	U

